MP Briefing — BioRevolution Coalition

Backing the Bioeconomy for the UK's Future Growth and Resilience.

About this briefing

The BioRevolution Coalition is a nation-wide campaign convened by BBIA — the Bio-based and Biodegradable Industries Association, the UK trade body for the bioeconomy — to unite citizens, industry and parliamentarians behind a coordinated UK policy review. “Bio-based Solutions” is the term used throughout for everyday products made from renewable biological resources rather than fossil oil and gas. This campaign concerns industrial bio-based chemicals and materials, not human-health medicines. For more on this please visit **biorevolution.uk**.

At a glance

The UK’s Modern Industrial Strategy envisions the nation as one of the world’s top three hubs for technology driven innovation by 2035, anchored by the ambition to create the UK’s first trillion-dollar tech company. Engineering Biology, and the wider bioeconomy — the cornerstone of this new industrial era and one of the Industrial Strategy’s 8 growth-driving sectors — drives the sustainable design and manufacture of fuels, chemicals, materials, food, and medicines. Globally valued at over £30 trillion by 2050, the bioeconomy represents a transformative opportunity for the UK.

UK-designed and manufactured bio-based chemicals and materials offer a major opportunity to **secure industries as we transition away from oil and gas**. Building on world-leading expertise in these sustainable technologies, the UK can drive net zero while creating and protecting **hundreds of thousands of highly skilled jobs**.

Chemicals are essential to everyday life, underpinning plastics, food, textiles, energy, batteries, defence, electronics, and medicines. They support food security, communications, healthcare, home heating, and national security.

Today, most chemicals are produced from fossil fuels and account for **around 10% of global greenhouse gas emissions**. Bio-based chemicals and materials use renewable biological resources — including plants, algae, mycelium, and organic waste — reducing reliance on fossil oil and gas.

The government has already committed **£2 billion over the next decade**, yet **fewer than 1 in 10 of the bio-based innovations it funds make it out of the lab**. The science is proven and the technology is deployable now. The bottleneck is policy: tax, regulation and waste rules that, taken together, leave bio-based alternatives significantly more expensive than the fossil-based incumbents they could replace. The cost so far – each year, **over 4,000 skilled UK jobs lost and £300–500m a year in GVA foregone**.

Despite this potential, several barriers are constraining progress in the UK. Bio-based alternatives often face higher costs than petrochemical incumbents, while limited domestic feedstock capacity reduces supply chain resilience. Most critically, regulatory complexity — particularly within UK REACH and related frameworks — creates significant delays, duplication and cost burdens, disproportionately affecting innovative bio-based products, which can be around 45% slower and twice as expensive to bring to market. These barriers slow commercialisation, deter investment and risk the UK falling behind global competitors. The evidence base for this briefing draws on BBIA's recent policy and finance reports, including its analysis of sector finance and investment, its Sustainable Materials policy paper, and its review of UK fermentation capacity.

Six facts to anchor any conversation:

01 This briefing concerns industrial bio-based chemicals and materials — the everyday products of the bioeconomy, made today mostly from fossil oil and gas and responsible for around 10% of global greenhouse gas emissions. It is not about medicines or human-health manufacturing, which are outside the scope of this campaign.

02 The UK Government has committed £2 billion over ten years to engineering biology and the wider bioeconomy. Between 2018 and 2024, over £450 million of UKRI funding went to bio-based research and innovation. Yet, fewer than 10% of those publicly funded innovations have reached commercial production in the UK.

03 The UK bio-based sector already supports more than 1,200 businesses (around 1,100 of which are SMEs), generated £12.5bn in revenue and £5.1bn in GVA in 2024, and supports a workforce of 20,300 employees.

04 Switching just 30% of UK chemical industry feedstock to biomass could generate up to £204bn in annual UK plc revenue and save more than 5.2 million tonnes CO_2eq each year — more than the Renewable Transport Fuel Obligation delivered in 2021.

05 Manufacturing bio-based chemicals and materials currently costs at least twice (and in some applications up to seven times) the equivalent fossil-based product. Current UK tax and regulatory frameworks widen that gap rather than narrow it.

06 More than half of bio-based companies interviewed by BB-REG-NET said they are considering scaling production overseas — principally in the EU or US — to access more stable regulatory and fiscal frameworks.

> The campaign proposition is much more than an environmental ask. It is an industrial policy and national resilience ask, with measurable returns to the Exchequer, regional jobs, and energy security.

1. What is the bioeconomy and why are Bio-based Solutions essential for UK's Future Growth and Resilience?

Most carbon used in everyday products today — plastics, textiles, cosmetics, cleaning products, fuels, medicines — originates from fossil oil and gas. Manufactured chemicals are responsible for around 10% of global greenhouse gas emissions.

The bioeconomy uses renewable biological resources — plants, microorganisms, agricultural and food wastes — as the carbon feedstock instead. The end products can be chemically identical to their fossil counterparts (so-called "drop-in" alternatives, for example bio-based versions of the polyethylene and PET used in everyday bottles and packaging), or chemically novel materials such as polylactic acid (PLA) or fermentation-derived ingredients.

Bio-based Solutions are everyday products made from renewable biological resources such as plants, microbes, and agricultural waste, rather than fossil resources like oil, coal, or gas. In simple terms, they use **carbon from nature instead of carbon extracted from underground**. Many Bio-based Solutions are created using engineering biology, where scientists use microorganisms to produce useful products without relying on fossil resources.

These products can be made from crops, wood and forestry waste, food and agricultural waste, microorganisms such as bacteria, yeast and algae, as well as seaweed and other marine materials. Many people already encounter Bio-based Solutions in daily life without realising it. Examples include alternative proteins made through fermentation, plant-based foods and cultivated meat; plastics derived from plants such as corn or sugarcane; chemicals produced through natural fermentation processes similar to brewing beer; sustainable fabrics made from wood pulp or crop waste; and plant-based ingredients used in cosmetics and cleaning products.

> Bio-based Solutions matter because they can reduce carbon emissions by lowering dependence on fossil resources, strengthen UK supply chain resilience, create thousands of skilled jobs, support rural and industrial growth, and help deliver Net Zero.

2. Why this matters now

Economic growth and the Industrial Strategy

Engineering biology is named as one of the UK's six frontier technologies in the Modern Industrial Strategy (June 2025), within which £380 million of the £2 billion is allocated to engineering biology infrastructure and innovation. Yet the House of Lords Science and Technology Committee, in its January 2025 report *Don't Fail to Scale*, found the UK to be science-rich but commercially under-developed and warned it is in severe danger of losing its lead in engineering biology — a textbook scale-up gap between excellent research and deployed industry.

Energy and supply chain resilience

The UK imports the overwhelming majority of the fossil carbon that goes into its chemicals and materials – making its cost heavily reliant on geopolitical events, supply-chain disruption and price shocks. The 2022 Russian invasion of Ukraine alone is estimated to have cost the UK £183 billion over four years through energy price effects — roughly equivalent to the entire cost of the net zero transition to 2050. Renewable, domestic carbon feedstocks are a hedge against precisely that exposure.

The UK can already grow or recover plenty of renewable feedstock at home. Examples include energy crops such as miscanthus and willow, agricultural sources like sugar beet and wheat straw, forestry and wood-processing residues, and food and farm waste. The right policy signal would let this domestic supply scale up.

Net Zero

Replacing just 12 high-potential fossil-based chemicals with bio-based equivalents could save more than 5.2 million tonnes CO_2eq annually — greater than the GHG savings achieved by the Road Traffic Fuel Obligation in 2021. Bio-based plastics in NHS supply chains alone could deliver 498 ktCO_2e in carbon savings, on the NHS's own modelling.

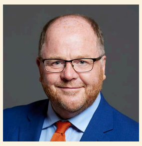
International competition

The UK is the only G7 nation without a current national bioeconomy strategy. Across Europe, 22 countries have either adopted or are developing dedicated bioeconomy strategies, anchored in the EU Bioeconomy Strategy and Circular Economy Action Plan. The US BioPreferred Program mandates federal procurement preference for over 7,000 certified bio-based products. Japan is targeting a 100 trillion-yen (£500 billion) bioeconomy market by 2030 with clear ministerial leadership in the Cabinet Office. The UK, by contrast, withdrew its 2018 *Growing the Bioeconomy* strategy in 2021 and has not replaced it.

3. The Coalition petition and its cross-party backing

In response, BBIA has convened the UK BioRevolution Coalition a new nation-wide campaign aimed at uniting citizens, industry, and parliamentarians with a mission to accelerate the UK's bioeconomy. We have launched a public petition on **petition.parliament.uk** calling for a coordinated policy review to level the playing field for bio-based solutions. Alongside this, we have convened hundreds of organisations, creating the UK's largest single voice for the bio-based sector, and we are working on creating an All-Party Parliamentary Group for the Bioeconomy, which currently has support across the House from George Freeman MP (Conservative), and Rachael Maskell MP (Labour).

Cross-party champions in the House:



George Freeman MP
(CONSERVATIVE)

Rachael Maskell MP
(LABOUR)

The petition is supported by a coalition of industry, trade bodies and research organisations — and is led by the BBIA – the UK's trade association for the bioeconomy.

4. Where the system is failing — five barriers

The barriers set out below are drawn from extensive research into the sector, including the work of the BBIA-convened BB-REG-NET regulatory network. They fall into five interlocking domains. Each is followed by questions for the Government which surface naturally from the evidence and remain — to date — unanswered.

Barrier 1 — Strategic vacuum: no plan, no dedicated minister, no accountability

The UK's *Growing the Bioeconomy* 2018 strategy, co-authored by BEIS and Defra, was quietly withdrawn in 2021. Since BEIS was dissolved, accountability for the bioeconomy has fragmented across four departments at minimum — Defra (environment, EPR, waste), DESNZ (biomass, ETS), DSIT (engineering biology, R&D) and HM Treasury (PPT, fiscal levers) — with the Department for Transport adding a fifth where biofuels are concerned. No single minister is accountable for the cross-cutting industrial outcome. Initiatives have proliferated without coordination or evaluation.

By contrast: the US Inflation Reduction Act and Bioeconomy programme is led from the Executive Office of the President; the EU bioeconomy is a central pillar of the European Green Deal; Japan's strategy is owned by the Integrated Innovation Strategy Promotion Council under the Cabinet Office.

OUTSTANDING QUESTIONS FOR GOVERNMENT:

- DSIT currently leads on engineering biology, but which single minister holds end-to-end accountability for delivering the Government's £2 billion engineering biology and bioeconomy commitment across all the departments involved — Defra, DESNZ, DSIT, HM Treasury and the Department for Transport?
- When will the forthcoming Circular Economy Growth Plan (expected early 2026) explicitly address bio-based and biodegradable materials, and how will it interlock with the Modern Industrial Strategy and the National Materials Innovation Strategy? The Coalition's case studies viewable at biorevolution.uk — including how current policy slows bio-based packaging reaching the market — show what is at stake if these strategies are not joined up.
- Will the Government commit to publishing a successor to the *Growing the Bioeconomy* strategy, with a designated Minister of State, comparable to the arrangements for net zero and space?
- What evaluation framework is in place to measure whether the £2 billion commitment is delivering commercialisation, and not only research outputs?

Barrier 2 — Tax: bio-based and fossil-virgin treated as equivalents

The UK's environmental tax structure — principally the Plastic Packaging Tax (PPT, £225/tonne) and Extended Producer Responsibility (EPR) — was designed to drive recycled content into fossil plastics. It has had unintended consequences for bio-based and compostable materials.

The PPT applies the same per-tonne rate to bio-based plastics as to virgin fossil plastics. Plastics with over 30% recycled fossil content are exempt. Plastics designed to biodegrade are not exempt. The result: a bio-based, compostable bag pays the same tax as a fossil-based equivalent, while a fossil-based bottle with recycled content does not.

Defra's Recyclability Assessment Methodology (RAM) under EPR places compostable and biodegradable materials in the "other" / RED band — "hard to recycle" — triggering approximately 20% higher fees. RAM evaluates only end-of-life recyclability; it makes no reference to feedstock and does not recognise the carbon benefit of bio-based content.

The Emissions Trading Scheme (ETS) recognises biogenic carbon, but the benefit is insufficient to offset the penalties imposed by EPR, RAM and the PPT.

Defra is not due to review the RAM in respect of bio-based and compostable materials until 2029 — widely viewed by the sector as too late, given investor confidence is already eroding and companies are already relocating.

This is the policy trap: the materials best placed to reduce environmental harm are also the least commercially viable under current rules.

OUTSTANDING QUESTIONS FOR GOVERNMENT:

- Why are bio-based plastics treated identically to virgin fossil plastics under the Plastic Packaging Tax, when they are made from renewable, non-fossil feedstocks and their lifecycle carbon footprint is materially lower?
- Will HMT bring forward the review of the PPT under section 49(8) of the Finance Act 2021, which permits revision without primary legislation?
- Will Defra accelerate the RAM review of bioplastics and compostables to no later than 2026, and create distinct EPR categories for compostable and natural polymer materials?
- Will HM Treasury commission a coordinated review of how the ETS, EPR and PPT together treat biogenic content — with a view to consistent recognition across all three frameworks?
- What is HMT's assessment of the case for targeted tariff exemptions on imported bio-based feedstocks, given UK manufacturers' reliance on biomass inputs not yet produced at scale domestically?

Barrier 3 — Regulation: built for fossil-based incumbents, slow for everything else

UK regulation is theoretically feedstock-neutral. In practice, the Coalition's *Bio-Barometer Survey* of over 100 sector stakeholders rated the UK's bio-based policy environment 2.49 out of 5. Regulation was identified as the single largest barrier to commercialisation — ahead of policy, standards and certification.

Specific issues:

- Inconsistent definitions of "plastic" across the Single-Use Plastics Ban, the Plastic Packaging Tax and UK REACH. The same bio-based material can be a "plastic" under one regulation and exempt under another. Crucially, none of these regulations defines what "chemically modified" means, leaving natural polymers (starch blends, chitosan, plant-protein based products) in a permanent grey zone.
- Approval of bio-based chemicals through UK REACH typically takes roughly twice as long and costs roughly twice as much as the equivalent fossil-based product, with no risk-proportionate fast-track for renewable-carbon materials.
- BB-REG-NET case study evidence shows that for just three bio-based SMEs, the UK is losing £35.7 million in annual GVA and over 473 unrealised jobs. Extrapolated across hundreds of similar SMEs, the avoidable annual loss is in the tens of billions.
- The Government's own Modern Industrial Strategy commits to a 25% reduction in regulatory administrative costs for businesses by the end of this Parliament. There is no equivalent commitment, scope or mechanism specific to bio-based products.

OUTSTANDING QUESTIONS FOR GOVERNMENT:

- Will Government establish a single, consistent definition of "plastic" across UK regulation that distinguishes fossil-derived, bio-based, and unmodified natural polymers?
- What progress has DSIT's Regulatory Innovation Office made on a co-developed regulatory roadmap for novel bio-based cosmetic ingredients, promised to the sector in November 2025?
- How will the 25% regulatory cost-reduction commitment in the Industrial Strategy apply specifically to bio-based products, given they currently face roughly twice the approval cost of fossil incumbents?
- What is the timeline for adopting BS EN 18027:2025 — the new bio-based products LCA standard — across UK regulatory and procurement frameworks, replacing legacy standards built around fossil-based incumbents?

Barrier 4 — Procurement: a market signal the UK is not using

Public procurement is one of the fastest, lowest-fiscal-cost levers a government has for creating early markets in emerging green industries. It is one of the levers the UK is using least.

The NHS is one of the largest single public purchasers in the UK, spending an estimated £8 billion annually on medical equipment and consumables. BB-REG-NET analysis of public contract data identified 670 NHS contracts for single-use plastic products worth £6.7 billion, of which 60% had high potential for bio-based substitution.

An NHS Supply Chain catalogue of approximately 500,000 products yielded just one result when searched for “bio-based” materials.

For just two further public-sector categories — catering disposables and refuse sacks — BB-REG-NET identified £177 million in potential annual demand across local authorities, central government and housing associations.

Internationally: the US BioPreferred Program mandates federal procurement of over 7,000 certified bio-based products. The EU Green Public Procurement Framework sets renewable-content criteria; Italy and the Netherlands mandate bio-based thresholds in public contracts. France’s RE2020 requires 50% bio-based content in new buildings. The UK has no equivalent.

The 2023 Procurement Act gives UK contracting authorities the legal flexibility to set such criteria. The political signal to use it has not yet been given.

OUTSTANDING QUESTIONS FOR GOVERNMENT:

- Will the Department of Health and Social Care commit to a pilot NHS bio-based procurement programme for high-volume single-use products (gowns, gloves, drapes, catering disposables, patient slings)?
- Will the Cabinet Office and Crown Commercial Service mandate compostable food service ware in national procurement frameworks, particularly in closed-loop settings (NHS, prisons, festivals, sporting events) where dedicated organic waste collection can be guaranteed?
- For medically contaminated waste destined for high-temperature incineration, which represents around 156,000 tonnes annually in NHS England alone, will Government mandate the use of bio-based materials in those product categories — so that the carbon released is biogenic and not fossil?
- What review is the Government undertaking of bio-based opportunities in the built environment, given construction is responsible for 25% of UK emissions and the Government has committed to delivering 1.5 million new homes?

Barrier 5 — Finance: the missing middle between R&D and commercial scale

The UK is good at funding bio-based research and bad at funding bio-based scale-up. The sector calls this the "missing middle" — the gap between Innovate UK demonstration grants and commercial production capacity.

Of more than £450m UKRI funding 2018–2024, fewer than 10% of funded bio-based innovations reached UK commercial production.

BB-REG-NET's *Bio-Barometer Survey* found 72% of respondents would expand UK operations if long-term financing or off-take guarantees were available.

Venture capital in the sector is dominated by early-stage seed funding; Series B and C rounds are routinely raised overseas.

BB-REG-NET projects that under a "scale-up infrastructure investment" scenario, the bio-based sector could grow at +18.6% CAGR to 2030, versus –5.1% under business-as-usual. The gap between the two scenarios is approximately £0.5 billion in foregone annual revenue.

There is no UK equivalent to the production tax credits in the US Inflation Reduction Act or the patient infrastructure capital deployed under the EU's Green Deal Industrial Plan.

OUTSTANDING QUESTIONS FOR GOVERNMENT:

- Will the National Wealth Fund or the British Business Bank establish a dedicated scale-up facility for Bio-based Solutions — bio-based chemicals and materials, modelled on the offshore-wind energy-transition funding mechanisms that closed cost-parity in that sector?
- How will the £184 million Engineering Biology Scale-up Infrastructure Programme be prioritised toward first-of-a-kind pilot and demonstration plants for bio-based products, rather than further research facilities?
- Will HMT extend R&D tax credits to include scale-up activities, not only research and development, to support the transition from lab to commercial deployment?
- Will the Government commission research into how the Renewable Transport Fuel Obligation (RTFO) model — already proven to drive private investment in low-carbon fuels — could be adapted for non-fuel bio-based products, including a possible Low Carbon Chemicals and Materials Mandate analogous to California's Low Carbon Fuel Standard?

5. The five-point ask

The Coalition's policy ask is a compact, deliverable roadmap drawn from extensive research into the barriers facing the sector:

ASK 01

Name a government lead.

A single Minister of State with end-to-end accountability for the bioeconomy, mirroring the arrangements for net zero and space.

ASK 02

Fix the policy and tax mismatches.

Ensure policy, regulation and taxation account for where a product's carbon comes from (its feedstock) and what happens to it at end of life. Bring forward the RAM review to 2026; reform the Plastic Packaging Tax to recognise bio-based content and exempt biodegradable materials intended for organic recycling; create a proportionate, faster UK REACH pathway for renewable-carbon materials; and commission an HMT-led alignment review across ETS, EPR and PPT. The forthcoming Biomass Strategy should also be widened: at present it focuses on burning biomass for energy rather than using it for higher-value chemicals and materials.

ASK 03

Align carbon accounting.

Adopt BS EN 18027:2025 as the standard for bio-based vs fossil-based comparative life cycle assessment in regulation and procurement, and develop the additional impact categories (biogenic carbon, biodegradability, microplastic formation) needed for fair comparison.

ASK 04

Use public procurement.

Use government buying power to create market demand: a guaranteed early market lets Bio-based Solutions scale, which is what brings their cost down. Establish a UK Bio-Preferred Procurement Scheme, beginning with an NHS pilot in high-volume single-use product categories, alongside Cabinet Office mandates for compostable food service ware in closed-loop settings.

ASK 05

Back first-of-a-kind plants.

Co-funded scale-up capital through the National Wealth Fund and British Business Bank, prioritised use of the £184 million Engineering Biology Scale-up Infrastructure Programme for first-of-a-kind plants, scale-up tax credits, and an exploration of an RTFO-style mandate for industrial chemicals and materials as a fiscal-policy lever, alongside the tax reforms in point 2.

6. Quick reference — the numbers

FIGURE	SOURCE
£2bn UK Government commitment to engineering biology and the wider bioeconomy	UK Modern Industrial Strategy (2025)
£450m+ UKRI funding to bio-based research and demonstration, 2018–2024	BB-REG-NET, From Research to Revenue
<10% of UKRI-funded bio-based innovations reaching UK commercial production	BB-REG-NET, From Research to Revenue
1,200+ businesses, ~1,100 SMEs in the sector	Perspective Economics, 2025
£12.5bn revenue, £5.1bn GVA, 20,300 employees in 2024	Perspective Economics, 2025
£204bn potential annual UK plc revenue from 30% chemicals feedstock switch	DESNZ-commissioned study (RAF097/2223)
5.2m+ tonnes CO_2eq annual saving from substituting just 12 high-potential bio-based chemicals	DESNZ RAF097/2223; BB-REG-NET
4,500 new skilled jobs annually, achievable with reformed UK policy	BB-REG-NET, 2025
£225/tonne Plastic Packaging Tax applied uniformly to virgin fossil and bio-based plastics	HMRC
670 NHS contracts (£6.7bn) for single-use plastics, ~60% suitable for bio-based substitution	BB-REG-NET, Buying into Biomanufacturing
£177m additional annual public-sector demand identified in catering disposables and refuse sacks alone	BB-REG-NET
£183bn UK economic cost of fossil price shock following Russia's invasion of Ukraine, 2022–26	Cited in BB-REG-NET, Modern Industrial Bioeconomy white paper
£35.7m annual GVA, 473 jobs lost from just 3 UK bio-based SMEs unable to scale	BB-REG-NET, 2025
156,000 tonnes annual NHS clinical waste (England) destined for high-temperature incineration	NHS England

7. Further reading

BBIA — the Bio-based and Biodegradable Industries Association — is the UK trade body for the bioeconomy and the convenor of the BioRevolution Coalition. BB-REG-NET, the Bio-based and Biodegradable Materials Regulatory Network, is the BBIA-convened research network whose evidence base underpins this briefing. The full BB-REG-NET white paper *Growing the UK's Modern Industrial Bioeconomy: Driving Economic Growth with Bio-based Solutions for People and Planet* (foreword by Alistair Carmichael MP, Chair of the EFRA Select Committee) underpins this briefing and contains the supporting evidence in full. Available on request.

Key BBIA reports:

- BBIA — Bioeconomy Finance and Investment
- BBIA — Sustainable Materials Policy Paper
- BBIA — UK Fermentation Capacity (covering bio-based cosmetic ingredients)

BB-REG-NET reports referenced:

- Plastic Definitions in UK Regulation
- From Research to Revenue
- Buying into Biomanufacturing: Harnessing Public Procurement for the UK Bioeconomy
- Standardised but Unfair (LCA standards review)
- Beyond the Label: UK Public Attitudes and Policy Gaps
- Bio-Barometer Survey
- Do Biodegradable Plastics Encourage Littering?
- Addressing Persistent Plastic Pollution: The Case for Biodegradable Solutions

Coalition contact

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A printable version of this briefing, suggested wording for written parliamentary questions, and constituency-level data can be made available on request. The Coalition would welcome the opportunity to brief Members and committees in person.